Band-1 Hadamard Order 9312 – Classical Paley-I (q=9311); IN SPK via Paley-I; Gap 2052 Still OPEN

H_9312 Paley-I PASS; Paley-II SKIP (q=4655=5*7^2*19 NOT prime power); Kronecker FAIL/SKIP (4*2328; 2328===0 mod 4 admissible but no on-box H_2328); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-21

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
Parents. `2026-09-21_Band1_Hadamard_9204_PaleyI`; `2026-09-21_Band1_Hadamard_9200_PaleyI`; `2026-09-21_Band1_Hadamard_9188_PaleyI`.
Lab. `/workspace/experiments/band1/hadamard_9312_paleyI/`.
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Honesty stamp (non-negotiable).
This memo claims **finite construction of order 9312 only (verified max|HH^T-nI|=0), via classical Paley-I q=9311 (prime ===3 mod 4). 9312 IN SPK via Paley-I.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 9312: Paley-I **PASS**s ($q = 9311$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL** ($q = 4655 = 5 \cdot 7^2 \cdot 19$ NOT prime power); Kronecker **FAIL/SKIP** (note $9312 = 4 \cdot 2328$; $2328 \equiv 0 \pmod 4$ admissible but no on-box H_2328). **9312** folds into SPK **only** via classical Paley-I (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **9204+9200+9188+9152+9104+9128+9124+9092+2220+8884+2236+2160+2140+8564 / 1064+2128 / ...** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

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Abstract

Finite construction of order **9312**.

- H_{9312} = Paley-I ($q = 9311$) – classical SPK
- Paley-II for 9312: **SKIP** ($q = 4655 = 5 \cdot 7^2 \cdot 19$ NOT prime power)
- Kronecker for 9312: **FAIL/SKIP** ($4 \cdot 2328$; $2328 \equiv 0 \pmod 4$ admissible but no on-box H_2328)

- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-I.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

```
python3 verify_hadamard_9312.py
```

Results. overall_pass=True; units 117/117; wall 656.2762s; constructed max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 9312 = q + 1$ with $q = 9311$ prime $\equiv 3 \pmod 4$ -> Paley-I eligible.
- Paley-II: $q = 4655 = 5 \cdot 7^2 \cdot 19$ NOT prime power -> SKIP.
- Kronecker: $9312 = 4 \cdot 2328$; $2328 \equiv 0 \pmod 4$ admissible but no on-box H_2328 – FAIL/SKIP.
- Miyamoto digression skipped this wave (prefer classical SPK).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_9312 Paley-I	Paley-I q=9311	0	648.4367
H_9312 Paley-II	SKIP q=4655=5*7ˆ2*19 NOT prime power	–	–
H_9312 Kronecker	FAIL/SKIP (4*2328; 2328===0 mod 4 admissible but no on-box H_2328)	–	–

SPK closure note

Seeds include prior SPK through **9284** lineage (**9284**; **9240**; **9228**; **9204**; **9200**; **9188**; **9152**; **9104**; **9128**; **9124**; **9092**; **2220+8884**; **2236**; **2160**; **2140+8564**; **1064+2128**; . . . ; **NOT** 8948/8644/8516/. . .) + verified **9312**. Next constructive gap **2052** (OPEN).

Quantity	Value
9312 in SPK	**True** (Paley-I q=9311)
Gap 2052	**OPEN**

Quantity	Value
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

Next optional

q=9319->n=9320 Paley-I DIRECT q=9319 OR q=9323->n=9324 Paley-I DIRECT q=9323 OR n=2052 standard SPK constructive gap – method-attack / lit OR q=9343->n=9344 Paley-I DIRECT q=9343 OR method-attack SPK gap 2052 OR Track B / Object 1 meter await